Test 1: Accelerometer vs Piezo sensor

Perform the testing twice. Once using a piezoelectric sensor, and the other using an accelerometer.

1. Setup
    a. Construct the circuit using a microcontroller (programmed using Arduino), with desired sensor, and a breadboard
    b. Obtain a ball of fixed mass and size (ideally a rubber ball), and a stable table
    c. Place a sound source that can produce at least 90 dB of noise on the table (simulate Stryke's background noise). Confirm the noise is approximately 90 dB using *Sound Meter*, a Google Play phone app.
    d. Calibrate the sensitivity so that the sensor does not register the background noise
    e. Place the sensor so that it is pressed up below the surface of the table
2. Testing
    a. Drop the ball from an initial height where the sensor cannot register force of the ball hit (measuring the shortest distance from the table's surface to the bottom of the ball). Record the initial height.
    b. Increase the height in 5 cm increment, and repeat until the hit is registered. Record the height where the sensor first records a hit.
3. Analysis
    a. The sensor that can detect the ball's hit from a lower height is deemed to be more suitable as our sensing method.

Note: For more consistent results, when using different sensors to compare results, the same microcontroller, table (including where the sensor attaches to the table), the ball (including the location where it hits the table), and the breadboard must be used.

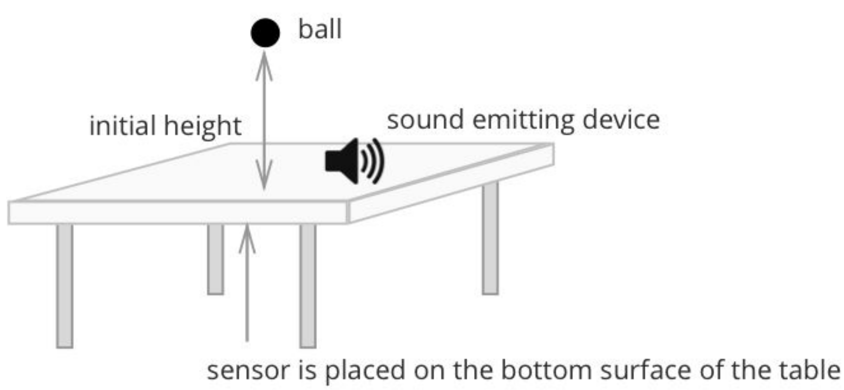
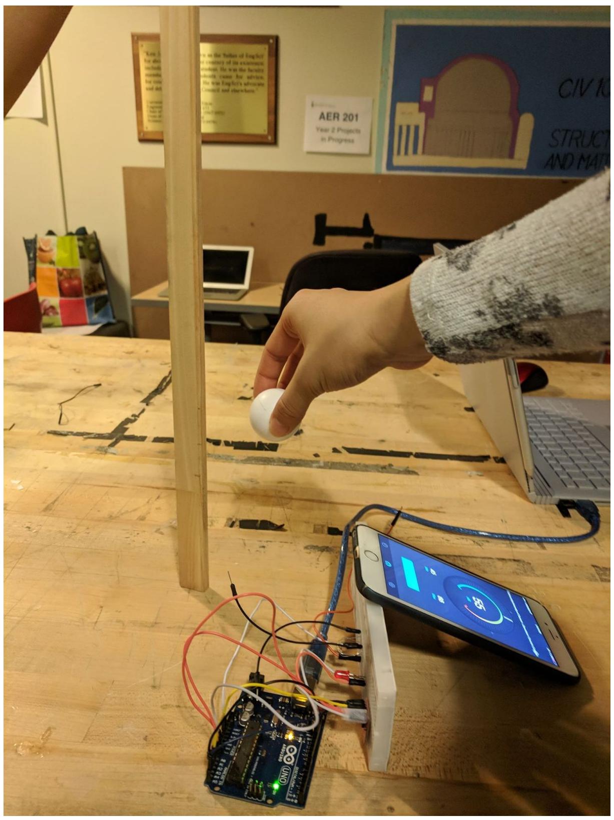
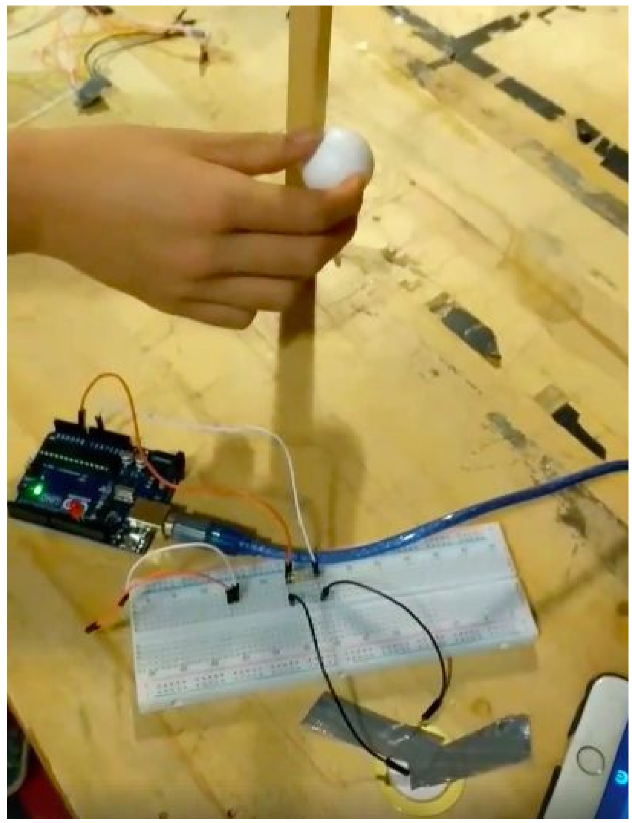


Figure 1: Setup of the experiment.

Results from Test 1

The test was set up and performed in the EngSci common room using a wooden table found in the kitchen. Using an app on our cellphones, the background noise was measured to be between 86 dB - 92 dB, which was deemed appropriate for background noise. A foosball ball from the common room was used for testing.





Figures 1 and 2: Initial position with the accelerometer and piezoelectric sensor.

The sensors were placed on the surface of the table rather than below it due to the delicacy of the wires; they were not soldered and hanging them upside would break apart the circuit.

Following the procedure for each sensor, the ball was dropped at a low height in close proximity to the sensor, and until the drop was registered, the height incrementally increased by 5cm. The following results were obtained. The sensor was connected to a light that flashed when the drop was registered for easy detection.

	Accelerometer	**Piezoelectric Sensor**
Initial height where drop is registered.	72 cm	1 cm *
Success rate (out of 10 trials)	70%	100%

*This was the shortest possible distance from the table where we held it and could pick it back up once it bounced.

From the results, the initial height and success rate of the piezoelectric sensor outperformed the accelerometer. To verify the piezoelectric sensor, the ball was dropped at different locations on the table, and all drops were registered. Ultimately, the piezoelectric sensor was selected for our device.

Test 2: Probability distribution of where the knife hits the board

1. Set-up
    a. Using string, divide the area of the board into square (approx. 6 rows and 5 columns), as well as the area around the board.
    b. Prepare an Excel sheet with a cell for square
2. Testing
    a. Throw the knife 5 m from the board. Increment the count of each square when it gets hit in the Excel sheet.
    b. Throw 400 times.
3. Analysis
    a. Calculate the probability that the knife hits where the LTD will be located (row 1, column 4)

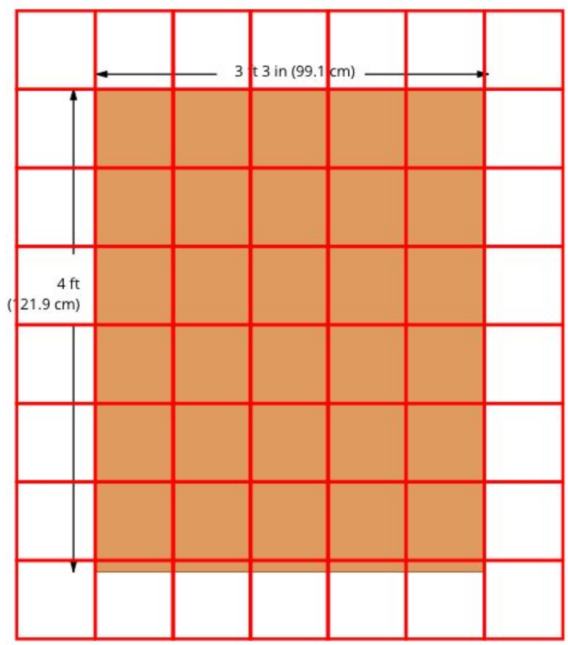


Figure 1: Setup of the experiment. Hanging string would be where the red lines are.

Results from Test 2

Figure 1: Set up of the experiment at Stryke Target Range.

0	0	0	1	0	0	0
0	0	0	3	4	0	0
0	0	3	25	22	2	0
0	1	14	61	45	10	1
0	2	13	86	34	5	0
0	0	10	27	17	1	0
0	0	0	9	3	0	0
0	1	0	2	0	0	0

Figure 2: The raw data obtained from our test. Yellow means the area is on the target and grey means it is the area outside the target.

0%	0%	0%	0.2%	0%	0%	0%
0%	0%	0%	0.7%	1.0%	0%	0%
0%	0%	0.7%	6.2%	5.5%	0.5%	0%
0%	0.2%	3.5%	15.2%	11.2%	2.5%	0.2%
0%	0.5%	3.2%	21.4%	8.5%	1.2%	0%
0%	0%	2.5%	6.7%	4.2%	0.2%	0%
0%	0%	0%	2.2%	0.7%	0%	0%
0%	0.2%	0%	0.5%	0%	0%	0%

Graded Color Scale

Figure 3: The probability of each section getting hit. Our raw data was divided by the total number of throws (402). Red means that the area has a higher chance of getting hit and green means that the area has a lower chance (in this case 0%) of being hit.

From our results, there was a 1/402 ≈ 0.2% chance that the area containing the LTD gets hit. It should be noted that beginners were used in this test (members of our team) and that in a professional/competitive setting, Humphrey, a professional knife thrower at Stryke, assures that the chances of players missing the board is even smaller than the results of our testing.

Our test shows that in the setting of beginners, perhaps most customers of Stryke, it is possible for the LTD to be hit, and there is a 0.2% chance. This further validates that we need to choose materials that can withstand large impacts.

Test 3: Material (Frontal Surface) Testing

1. Setups
    a. Prepare a sheet of the selected material, with the same dimension of that are required to build the frontal surface of the device.
    b. Adhere 2 small pieces of wood to two opposite edges of the sheet so that when attached to the board, the sheet is not in direct contact with the board.
    c. Secure such material to a fixed location of the board.
2. Testing
    a. Stand 5 meters away from the board and hit the material 10 times.
    b. Observe and record the damage done to the material

3. Analysis
    a. Assess the performance of the material.

Unacceptable	**Acceptable**	**Good**	**Ideal**
The material surface is broken through/shatters and can no longer hold its shape	The material surface is broken through but holds its shape/does not shatter	The material has dents/scratches and holds its shape	The material shows no indications of being struck

Note: the knife, external environment (temperature, wind), people participated in the testing, distance from target, thickness of the material, angle of the sheet comparing to the board shall remain the same when testing of every candidate material.

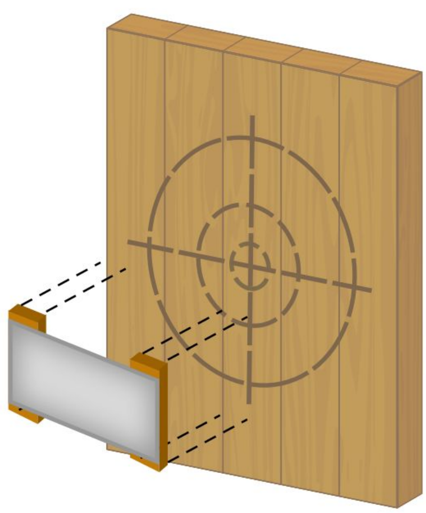


Figure 2: Setup of the experiment

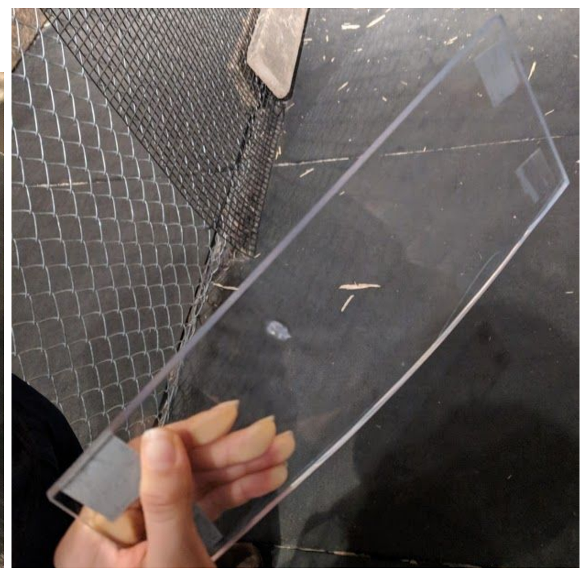
Results from Test 3

Due to the timeline of showcase and the time it takes for the materials to ship, we deemed it necessary to minimize the chances of failure so we decided to test the best material from our rating matrices. From our rating matrix, the best glass was polycarbonate, and the best metal was stainless steel.

Figure 1 and 2: How the materials were attached to the board. Wooden blocks were used to separate the board and the sheet, then everything was duct taped together.

The sheets were attached to the board with 2 wooden blocks in-between so that the two surfaces were not in direct contact. The materials were then struck 10 times and then the damage was assessed. Both the glass and the metal had very light scratches that were deemed **good**.

Figure 3: To test if the polycarbonate and metal would shatter, we directly adhered it to the target.



Figures 4 and 5: The knife went through the polycarbonate and metal, but they did not crack or shatter.

To test the worst case scenario, we tried to apply as much force as we could with a knife. The sheet was then adhered directly to the board so that the surfaces were in direct contact. Humphrey, a professional knife thrower at Stryke, was asked to aim at the sheet with as much force as he could. With both materials, the knife broke through the surface of the sheet, but held its shape and did not crack/shatter, which was deemed **acceptable** under extreme conditions.

Design decision that came out of this test

Although the glass only had light scratches, over prolonged used, it may hinder the visibility of the LED screen and lights behind it. We then changed the design of our box to allow for easy removal. We simply added two additional strips of metal to create a slot that can hold the glass.

As well, since the front of the box would get hit the most, we decided to make the entire frontal surface made of glass, since it is removable.